Learning-Based Approach for Atmospheric Compensation of VNIR Hyperspectral Data

Nicola Acito, *Member, IEEE*, Marco Diani, *Member, IEEE*, and Giovanni Corsini, *Member, IEEE*

***Abstract*— In this work, we deal with the problem of atmospheric compensation (AC) of hyperspectral data collected in the visible and near-infrared (VNIR) spectral range. We propose the "learning-based" approach which uses artificial intelligence algorithms to directly estimate the spectral reflectance from the observed at-sensor radiance image. It uses a parametric regressor whose parameters are learned by means of a strategy based on synthetic data. Such data are generated taking into account 1) the radiative transfer in the atmosphere; 2) the variability of the surface spectral reflectance; and 3) the effects of signal-dependent random noise and spectral miscalibration errors. According to this general framework, we propose a specific multilinear regressor that starting from the knowledge of the atmospheric visibility compensates the water absorption and provides the spectral reflectance of each pixel of the analyzed image. Furthermore, a specific image-based procedure is presented for visibility estimation. The experiment over simulated data is presented and discussed. The test on simulated data aims at showing the effectiveness of the proposed strategy in a completely controlled environment. Experiments are also carried out on three real hyperspectral images acquired by two hyperspectral sensors. The obtained results confirm the effectiveness of the proposed approach by comparing the retrieved reflectance spectra with *in-situ* measurements or with those obtained by using a well-known commercial AC software.**

***Index Terms*— Atmospheric compensation (AC), hyperspectral imagery, learning-based (LB) approach, machine learning.**

I. Introduction

HYPERSPECTRAL sensors (HSs) measure the spectrum of the observed pixel in a large number of contiguous and very narrow spectral channels allowing the exploitation of the physical and chemical composition of the observed surface through its spectral reflectance. In recent years, thanks to the development of new generation sensors on board of unmanned aerial vehicles (UAVs) and drones, HSs operating in the visible and near-infrared (VNIR) spectral range have gained the interest of the scientific community ([1]–[3]).

Most of the hyperspectral data applications exploit the spectral reflectance of the observed materials to fully understand the phenomenology of interest. Unfortunately, the spectral signature collected by HSs from aerial or satellite platforms (at-sensor spectral radiance) is altered by the spectrally dependent distortion effects due to the atmosphere in the radiative transfer path ([4]). Thus, the measured at-sensor radiance needs to be processed to remove the contributions of solar illumination and the atmospheric effects. Such process, generally referred to as atmospheric compensation (AC [5]–[10]), plays a crucial role in the exploitation of hyperspectral data. AC is a very challenging task because the effects of solar illumination and atmospheric distortion depend on several factors not directly measurable at the acquisition time.

In the VNIR spectral range, the atmospheric effects are mainly influenced by the following ([9], [10]): 1) the aerosol extinction; 2) the absorption of uniformly mixed gases (O_2 and O_3) that are relatively stable in concentration; and 3) the water vapor absorption. Aerosol extinction has a smooth spectral behavior that depends mainly on the particle size and distribution and on the visibility V [or equivalently on the aerosol optical Depth (AOD)]. It varies slowly in spatial dimension and can be considered stable in relatively small scenes such as those acquired by aerial sensors.

The water vapor absorption has a sharp behavior in specific spectral windows (mainly around 0.7, 0.8, 0.9, and 1.13 μm) and is determined by the column water vapor concentration (CWV) through the path of the downward and upward radiation. CWV plays a major role in the radiative transfer and is characterized by high spatial variability also in small monitored areas ([5], [10], [11]).

AC methods can be broadly grouped into two macroclasses ([10]): scene-based (SB) ([12]–[14]) and physics-based (PB) methods ([8], [15]–[18]). The major difference between the two classes is that PB relies on a specific model that accounts for the physics of the radiative transfer in the atmosphere [radiative transfer model (RTM)], whereas SB includes methods that are based on empirical approaches without reference to the physics of radiation propagation through the atmosphere.

The PB methods are in general the most accurate and reliable ones, because they allow the absolute spectral reflectance to be obtained without any *a priori* knowledge of the materials in the monitored scene. Furthermore, contrary to the SB methods, the PB methods address the spatial variability of CWV using estimation algorithms that work on a per-pixel basis. Popular PB methods such as the fast line-of-sight atmospheric analysis of spectral hypercubes (FLAASH, [16]) and the airborne atmospheric and topographic correction (ATCOR, [8])

Manuscript received June 8, 2020; revised July 7, 2020; accepted August 5, 2020. Date of publication August 25, 2020; date of current version April 22, 2021. *(Corresponding author: Nicola Acito.)*

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Digital Object Identifier 10.1109/TGRS.2020.3016094

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rely on a two-step procedure. In the first step, the RTM model parameters (V and CWV) are estimated from the analyzed image. V is generally estimated by analyzing regions of the image characterized by low spectral reflectance values as in the case of the dense dark vegetation algorithm (DDV, [20]). For CWV retrieval, several algorithms have been proposed in the past, for instance, those based on the differential absorption approach ([21]) and, more recently, those based on a deep convolutional neural network ([22]).

In the second step, the estimated parameters are used to obtain the spectral radiometric quantities for RTM inversion and reflectance retrieval. The operations in both the steps are performed by exploiting a specific radiative transfer code (MODTRAN [19] in most of the commercial products) that relates the atmospheric model parameters to the spectral radiometric quantities.

An alternative approach could be the "learning-based (LB)" one which uses artificial intelligence algorithms to directly estimate, from the at-sensor radiance, the spectral radiometric quantities of the RTM or, even, the spectral reflectance, accounting for the variability induced by one or more unknown parameters of the physical model. Such algorithms could be trained using synthetic data based on a physics RTM and generated by means of a specific radiative transfer code. The LB approach can be seen as the new frontier of AC and deserves the attention of the scientific community in light of the remarkable results obtained in various application fields.

Studies in this direction were made in the context of AC of longwave infrared (LWIR) hyperspectral images. In particular, in [23] exploiting the coupled subspace model, a machine LB approach for the estimation of atmospheric transmittance and upwelling radiance (RTM spectral radiometric quantities) was proposed. More recently, Westing *et al*. [24] have introduced a very interesting deep LB algorithm for the estimation of the RTM spectral radiometric quantities (transmittance, upwelling radiance, and downwelling radiance) directly from the acquired LWIR hyperspectral image.

In this work, we explore the possibility of applying the LB approach to VNIR hyperspectral data. In particular, we introduce a general framework for building LB approaches aimed at estimating the spectral reflectance directly from the observed at-sensor radiance image.

Therefore, according to the introduced framework, we derive a machine-learning-based regressor that retrieves spectral reflectance, in the realistic case of spatially variant CWV concentration. The proposed reflectance regressor compensates the absorption effects of the water vapor without estimating the CWV values in each pixel of the image. It adopts *a-priori* knowledge about the atmosphere and the acquisition geometry generally available in real application. The algorithm needs an estimate of the visibility V, which is obtained by an estimation procedure that exploits the presence of low reflectance data at specific wavelengths of the spectrum covered by the HS and a lookup table (LUT) derived by MODTRAN.

Training of the reflectance regressor is performed over hyperspectral data synthetized according to one of the most popular RTM ([10]) and using a simulation procedure exploiting MODTRAN and the availability of spectral reflectance libraries. The learning strategy is specifically designed to provide the algorithm with some degree of robustness to noise and spectral miscalibration errors.

This article is organized as follows. In Section II, we briefly summarize the RTM and define the assumptions made in the derivation of the proposed method. In Section III, we introduce the general framework for the LB approach to AC and we detail the specific algorithm proposed in this work. Section IV is dedicated to the description of the learning strategy adopted for algorithm training. In Section V, we describe the procedure adopted to obtain the visibility estimates. Finally, in Section VI, we discuss the performance of the proposed method on simulated data. We conclude this article with a proof of concept concerning the application of the LB AC technique to real images acquired by two different hyperspectral airborne sensors.

II. At-Sensor Radiance Model in the VNIR Spectral Range

In this section, with reference to the VNIR spectral range, we introduce the model adopted to account for the radiative transfer in the atmosphere (RTM).

Denoting with $\mathbf{L}\epsilon R^{N_B}$ the at-sensor radiance vector, where N_B is the number of sensor channels, $\mathbf{L}$ can be expressed as ([9], [10], [16])

$$\begin{aligned}&\mathbf{L}\left(\boldsymbol{\rho}, \boldsymbol{\rho}_{\mathbf{a}}, \boldsymbol{\Gamma}, V, \mathrm{CWV}\right)\\&\quad=\mathbf{L}_{\mathrm{GND}}\left(\boldsymbol{\rho}, \boldsymbol{\rho}_{\mathbf{a}}, \boldsymbol{\Gamma}, V, CWV\right)+\mathbf{L}_p(\boldsymbol{\Gamma}, V, \mathrm{CWV})\\&\mathbf{L}_{\mathrm{GND}}\left(\boldsymbol{\rho}, \boldsymbol{\rho}_{\mathbf{a}}, \Gamma, V, \mathrm{CWV}\right)\\&\quad=\frac{\mathbf{Ecos}(\vartheta)}{\pi}\left[\boldsymbol{\tau}_1(\boldsymbol{\Gamma}, V, \mathrm{CWV})\frac{\boldsymbol{\rho}}{1-\mathbf{S}(\boldsymbol{\Gamma}, V, \mathrm{CWV})\boldsymbol{\rho}_{\boldsymbol{a}}}\right.\\&\qquad\left.+\boldsymbol{\tau}_2(\boldsymbol{\Gamma}, V, \mathrm{CWV})\frac{\boldsymbol{\rho}_a}{1-\mathbf{S}(\boldsymbol{\Gamma}, V, \mathrm{CWV})\boldsymbol{\rho}_{\boldsymbol{a}}}\right]\end{aligned}\tag{1}$$

where $\mathbf{L}_{\mathrm{GND}}(\boldsymbol{\rho}, \boldsymbol{\rho}_{\boldsymbol{a}}, \Gamma, V, \mathrm{CWV})$ is the vector of the spectral radiance reaching the sensor after interaction with the ground surface (ground reflected radiance), and $\mathbf{L}_p(\boldsymbol{\Gamma}, V, \mathrm{CWV})$ is the intrinsic upwelling atmospheric spectral radiance vector. The ground reflected radiance depends on the observed surface spectral reflectance $\boldsymbol{\rho}$ and on the adjacent surface spectral reflectance $\boldsymbol{\rho}_a$. Furthermore, it depends on the acquisition conditions by means of the radiometric quantities $\mathbf{E}$, $\mathbf{L}_p, \boldsymbol{\tau}_1, \boldsymbol{\tau}_2,$ and$\mathbf{S}$. Specifically, $\mathbf{E}$ is the vector of the solar spectral irradiance at the top of the atmosphere. The term $\boldsymbol{\tau}_1(\boldsymbol{\Gamma}, V, \mathrm{CWV})$ accounts for the interaction of the surface material with the direct and diffuse solar radiation, while $\boldsymbol{\tau}_2(\boldsymbol{\Gamma}, V, \mathrm{CWV})$ accounts for the part of the solar radiation that, after the interaction with the materials in a neighborhood of the surface of interest, is scattered by the atmosphere in the sensor instantaneous field of view (IFOV). The contributions due to multiple scattering are addressed by the term $1-\mathbf{S}(\boldsymbol{\Gamma}, V, \mathrm{CWV})\boldsymbol{\rho}_a$ with $\mathbf{S}(\boldsymbol{\Gamma}, V, \mathrm{CWV})$ denoting the spectral spherical albedo of the atmosphere.

The radiometric quantities in (1) are related to the parameters $\boldsymbol{\Gamma}, V$, and CWV that specify the acquisition geometry,

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the effects of absorption by gases in the atmosphere, and scattering by molecules and aerosol, respectively. The acquisition geometry is generally determined by the sun zenith (ϑ) and azimuth angles, the sensor zenith and azimuth angles, the sensor altitude (h), and the terrain elevation. In the following, we assume that data are acquired from nadir view (likely in the case of aerial applications) and that the monitored scene is flat. Under these assumptions, ϑ and h are the relevant parameters for characterizing the acquisition geometry. They are generally known in that they can be inferred from flight operation planning, acquisition date, and time.

The aerosol effects depend mainly on the particle size and distribution and on the visibility V. The aerosol type depends on the geographical position of the monitored scene, on the seasonal characteristics at the acquisition time, and, in general, on the concentration of the various contents of the atmosphere (depending on natural phenomena and human activities in the area). They can be modeled with good accuracy using standard values included in MODTRAN. The parameter V cannot be modeled by standard values and needs to be estimated from the specific analyzed image.

The concentration of the gases in the atmosphere determines the absorption effects. In the VNIR, spectral range absorption is due to the oxygen, which affects very narrow portions of the electromagnetic spectrum, by the ozone, whose effects are well-modeled by standard concentration values, and mainly by the water vapor which is highly variable in time and space. It is characterized by the total column water CWV.

In (1), we explicitly separated the unknown parameter V, CWV that strictly depends on the specific monitored scene from the rest of the model parameters (considered as known, hereinafter) that are grouped in the vector $\mathbf{\Gamma}$.

It is worth noting that in (1) we assumed that the observed surfaces are Lambertian and does not account for the anisotropic reflective behavior of those surfaces. To consider such effects, more sophisticated models should be adopted that greatly increase the complexity of the RTM. Furthermore, it is important to remark that (1) does not model the radiance from shadowed surfaces.

According to most of the existing AC algorithms ([8], [9], [16]) we assume that the vector $\mathbf{L_a}$ of the spectral radiance from THE adjacent regions is available and modeled as

$$\begin{aligned}&\mathbf{L}_{\boldsymbol{a}}\big(\boldsymbol{\rho}_{\boldsymbol{a}}, \mathbf{\Gamma}, V, \mathrm{CWV}\big)\\&\quad= \mathbf{L}_{\mathrm{GND}}\big(\boldsymbol{\rho}_{\boldsymbol{a}}, \boldsymbol{\rho}_{\boldsymbol{a}}, \mathbf{\Gamma}, V, \mathrm{CWV}\big) + \mathbf{L}_p(\mathbf{\Gamma}, V, \mathrm{CWV})\\&\mathbf{L}_{\mathrm{GND}}\big(\boldsymbol{\rho}_{\boldsymbol{a}}, \boldsymbol{\rho}_{\boldsymbol{a}}, \mathbf{\Gamma}, V, \mathrm{CWV}\big)\\&\quad= \frac{\mathbf{Ecos}(\vartheta)}{\pi}\left[\boldsymbol{\tau}_1(\mathbf{\Gamma}, V, \mathrm{CWV})\frac{\boldsymbol{\rho}_{\boldsymbol{a}}}{1-\mathbf{S}(\mathbf{\Gamma}, V, \mathrm{CWV})\boldsymbol{\rho}_{\boldsymbol{a}}}\right.\\&\qquad\left.+\boldsymbol{\tau}_2(\mathbf{\Gamma}, V, \mathrm{CWV})\frac{\boldsymbol{\rho}_{\boldsymbol{a}}}{1-\mathbf{S}(\mathbf{\Gamma}, V, \mathrm{CWV})\boldsymbol{\rho}_{\boldsymbol{a}}}\right].\end{aligned} \tag{2}$$

In practice, $\mathbf{L_a}$ is determined by averaging THE surrounding pixels by a specific weighting kernel ([9], [25]) that accounts for the atmospheric point spread function (PSF, [26], [27]).

Equations (1) and (2) define the RTM we will refer to in this work.

In the rest of this article, to simplify the notation, we will omit the dependence of $\mathbf{L}$ and $\mathbf{L_a}$ on $\boldsymbol{\rho}$, $\boldsymbol{\rho}_a$, and $\mathbf{\Gamma}$. It will only be reintroduced in the cases where the role of the parameters needs to be explicit.

III. Learning-Based Approach

Most of THE PB methods rely on a general procedure that first estimates the unknown parameters in the RTM, then derives the radiometric quantities corresponding to those parameters, by means of a radiative transfer code, and, finally, inverts the model to obtain the spectral reflectance $\boldsymbol{\rho}$. In this work, we propose an LB approach that differs from the summarized general procedure because it bypasses some of its processing steps. Specifically, we propose a method to estimate $\boldsymbol{\rho}$ directly from the at-sensor radiance $\mathbf{L}$ of the observed pixel and that from the adjacent regions $\mathbf{L_a}$ regardless of the spatially varying values of CWV, and without the calculation of the radiometric quantities. The only atmospheric parameter to be estimated is V. In the following, we first introduce the general framework for the LB approach to AC problem. Then, we focus on the description of the method we propose for estimating $\boldsymbol{\rho}$, enclosing it in that general framework. In this phase, we assume the parameter V as known, postponing the description of the algorithm for estimating V to Section V.

The general idea behind the LB approach is to define a parametric transformation (regressor) that allows to obtain the estimate $\hat{\rho}$ of $\boldsymbol{\rho}$ from the predictors $\mathbf{L}(V, \mathrm{CWV})$ and $\mathbf{L_a}(V, \mathrm{CWV})$. The regressor is a parametric function, denoted as g, operating the following transformation:

$$\hat{\boldsymbol{\rho}} = g(\mathbf{L}, \mathbf{L_a}; \mathbf{W}, V) \tag{3}$$

where $\mathbf{W}$ denotes the vector of the regressor parameters. In (3), we have explicitly indicated the dependence of the regressor on the parameter V according to the assumption made.

The response of the regressor to the predictors $\mathbf{L}(V, \mathrm{CWV})$ and $\mathbf{L_a}(V, \mathrm{CWV})$ depends on the specific value of the parameters' vector $\mathbf{W}$ that must be determined through the learning process. Learning is performed by exploiting the availability of N_T training samples, that is, triplets made up of the predictors $\mathbf{L}_t(V, \mathrm{CWV})$ and $\mathbf{L}_{\mathbf{a},t}(V, \mathrm{CWV})$ and the corresponding reflectance vector $\boldsymbol{\rho}_t$ ($t = 1, \ldots, N_T$), and by minimizing a given loss function over those training samples. In formulas

$$\mathbf{W} = \arg\min_{\hat{\mathbf{W}}} \underset{t=1,\ldots,N_t}{\mathrm{loss}}\big[g\big(\mathbf{L}_t, \mathbf{L}_{\mathbf{a},t}; \tilde{W}, V\big), \boldsymbol{\rho}_t\big]. \tag{4}$$

It is worth noting that our idea is to perform the training process over synthetic data that are generated according to the RTM in (1) and (2), using MODTRAN and by exploiting the availability of a spectral reflectance library. The detailed description of the training strategies is reported in Section IV, while, here, we focus on the description of the adopted parametric regressor and loss function. However, before delving into the description of the regressor and the loss function, let us take another step in the model definition.

The smooth behavior of the spectral reflectance of most of the natural materials and man-made objects ([28], [29]) suggests that the vector $\boldsymbol{\rho}$ can be considered as lying into

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a subspace Φ having rank K lower than that of the original data space (N_B). Thus, denoting as $\mathbf{U}$ one of the possible N_BxK basis matrices of Φ, the generic vector $\boldsymbol{\rho}$ can modeled as ([30])

$$\boldsymbol{\rho} = \mathbf{U} \cdot \mathbf{c} \tag{5}$$

where $\mathbf{c}$ is the $K \times 1$ vector of the components of $\boldsymbol{\rho}$ on the basis $\mathbf{U}$. With the model in (5) and assuming the availability of the basis matrix $\mathbf{U}$, the AC problem can be reformulated as the problem of estimating the components' vector $\mathbf{c}$ rather than $\boldsymbol{\rho}$. Accordingly, the regressor in (3) is modified as

$$\begin{aligned} \hat{\mathbf{c}} &= g(\mathbf{L}, \mathbf{L_a}; \mathbf{W}, V) \\ \hat{\boldsymbol{\rho}} &= \mathbf{U} \cdot \hat{\mathbf{c}} \end{aligned} \tag{6}$$

where $\hat{c}$ is the estimate of $\mathbf{c}$. The mathematical formulation of the learning process is rewritten as

$$\begin{aligned} \mathbf{W} &= \arg\min_{\tilde{\mathbf{W}}} \underset{t=1,\ldots,N_t}{\text{loss}}\left[g\left(\mathbf{L}_t, \mathbf{L}_{\mathbf{a},t}; \tilde{W}, V\right), \mathbf{c}_t\right] \\ \mathbf{c}_t &= \mathbf{U}^{\#} \cdot \boldsymbol{\rho}_t \end{aligned} \tag{7}$$

where $\mathbf{c}_t$ is the components' vector of $\boldsymbol{\rho}_t$, and $\mathbf{U}^{\#}$ is the Moore–Penrose inverse of $\mathbf{U}$.

The rationale behind the use of a low-rank representation for $\boldsymbol{\rho}$ is that dealing with an LB approach, it is a good practice to reduce the number of features to be estimated so as to decrease the risk of overfitting. Furthermore, using a low-rank representation we also gain in terms of computational load in that we decrease the computational complexity of the regressor. Note that the basis matrix $\mathbf{U}$ can be determined using the numerous libraries of spectral reflectance samples currently available ([31]–[33]). The specific procedure we adopt to obtain $\mathbf{U}$ from a given collection of spectral reflectances $\mathbf{D}$ is based on the singular value decomposition (SVD, [39]) and is detailed in Appendix I.

In this work, we define the regressor g through a multiple linear transformation

$$\hat{\mathbf{c}}=\mathbf{W_1^T}(V) \cdot \mathbf{L}(V, \text{CWV}) + \mathbf{W_2^T}(V) \cdot \mathbf{L}_{\boldsymbol{a}}(V, \text{CWV}) + \mathbf{w_0}(V) \tag{8}$$

where ^{T} denotes the matrix transpose, $\mathbf{w_0}(V)$ is the $K \times 1$ offset vector, and $\mathbf{W_1}(V)$ and $\mathbf{W_2}(V)$ are the KxN_B matrices determining the dependence of $\hat{c}$ on $\mathbf{L}(V, \text{CWV})$ and $\mathbf{L_a}(V, \text{CWV})$, respectively. Note that, in (8), we keep explicit the dependence of the regressor on V.

Defining the $(2N_B + 1) \times 1$ vector $\mathbf{x}(V, \text{CWV}) = [\mathbf{L}^{\text{T}}(V, \text{CWV}), \mathbf{L}_{\mathbf{a}}^{\text{T}}(V, \text{CWV}), 1]^{\text{T}}$, the expression of the regressor can be written in a more compact form as

$$\hat{\mathbf{c}} = \mathbf{W}(V)^{\text{T}} \cdot \mathbf{x}(V, \text{CWV}) \tag{9}$$

where $\mathbf{W}(V)$ is the $(2N_B + 1) \times K$ matrix defined as $\mathbf{W}(V) = [\mathbf{W_1^T}(V), \mathbf{W_2^T}(V), \mathbf{w_0}(V)]^{\text{T}}$.

To learn the regressor parameters $\mathbf{W}(V)$, we make use of the following loss function:

$$\begin{aligned} &\underset{t=1,\ldots,N_t}{\text{loss}}\left[g\left(\mathbf{x_t}; \tilde{W}, V\right), \mathbf{c}_t\right] \\ &= \sum_{t=1}^{N_t} \frac{\left\|\mathbf{c}_t - \tilde{W}(V)^{\text{T}} \cdot \mathbf{x_t}(V, \text{CWV})\right\|^2}{\|\mathbf{c}_t\|^2} + \beta \left\|\tilde{W}(V)\right\|_F^2 \end{aligned} \tag{10}$$

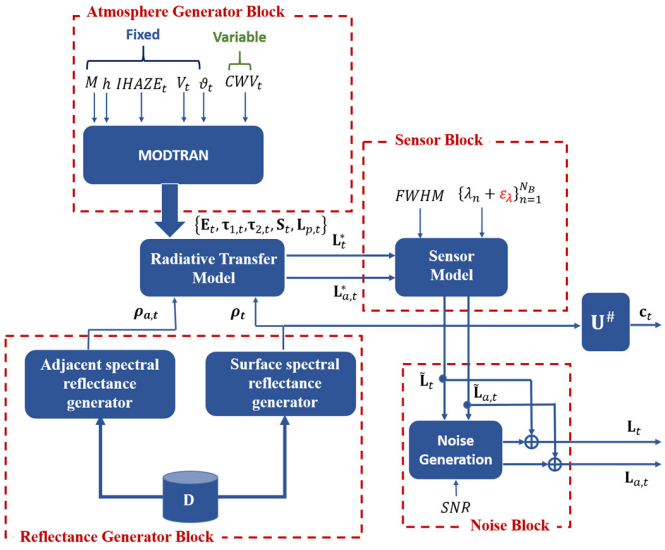


Fig. 1. Block diagram of the training set simulation strategy.

with $\mathbf{x_t}(V, \text{CWV}) = [\mathbf{L}_t^{\text{T}}(V, \text{CWV}), \mathbf{L}_{a,t}^{T}(V, \text{CWV}), 1]^{\text{T}}$. In (10), we denote $\|\|$ as the $L2$-norm and $\|\|_{\text{F}}$ as the Frobenius norm. The first term in the expression of the loss function is (by less than a multiplicative factor) the mean relative error between $\mathbf{c}_t$ and $\hat{c}_t$, while the second term is the $L2$ regularization introduced to limit the risk of overfitting and avoid ill-conditioning. The minimization problem in (7) with the loss function defined in (10) has the form of a weighted Ridge regression whose closed-form solution can be written as

$$\mathbf{W}(V) = \left[\sum_{t=1}^{N_t} \frac{\mathbf{x}_t(V, \text{CWV}) \cdot \mathbf{x}_t^{\text{T}}(V, \text{CWV})}{\|\mathbf{c}_t\|^2} + \beta \cdot \mathbf{I}\right]^{-1} \cdot \sum_{t=1}^{N_t} \frac{\mathbf{x}_t(V, \text{CWV}) \cdot \mathbf{c}_t^{\text{T}}}{\|\mathbf{c}_t\|^2}. \tag{11}$$

IV. Learning Strategy

The learning phase consists in deriving the regressor parameters $\mathbf{W}(V)$ in (11) using a training set of N_T simulated triplets $\{\mathbf{L}_t(V, \text{CWV}), \mathbf{L}_{\mathbf{a},t}(V, \text{CWV}), \mathbf{c}_t\}$ generated according to the RTM in (1) and (2). $\mathbf{L}_t(V, \text{CWV}), \mathbf{L}_{\mathbf{a},t}(V, \text{CWV})$ are the inputs of the model, while $\mathbf{c}_t$ is the output.

The simulation strategy is similar to that proposed in [22] and exploits MODTRAN to generate the radiometric quantities and a spectral reflectance library $\mathbf{D}$ for the terms $\boldsymbol{\rho}_t$ and $\boldsymbol{\rho}_{a,t}$. $\mathbf{D}$ contains the measured spectral reflectances of several man-made materials, vegetation, water, soils, and minerals and is obtained by merging the three available spectral libraries ASTER ([31]), USGS (version released in 2007, [32]), and ANGERS ([33]).

The simulation strategy (whose block diagram is reported in Fig. 1) has the four main blocks described in the following.

The Atmosphere Generator Block exploits MODTRAN to generate the radiometric quantities $\mathbf{E}_t$, $\boldsymbol{\tau}_{1,t}$, $\boldsymbol{\tau}_{2,t}$, $\mathbf{L}_{p,t}$, and $\mathbf{S}_t$ in the spectral range of interest (VNIR) and with high spectral

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TABLE I
LIST OF FIXED MODTRAN PARAMETERS ADOPTED FOR THE SIMULATION OF THE TRAINING SET FOR EACH REGRESSOR CONSIDERED IN THIS WORK

Fixed Atmospheric Parameters	
MODEL	1 (Tropical); 2 (Mid-Latidute Summer); 3 (Mid-Latitude Winter); 4 (Sub-Artic Summer); 5 (Sub-Artic Winter)
IHAZE	1 (Rural); 3 (Navy Maritime); 4 (Maritime); 5 (Urban); 10 (Desert)
$\vartheta(deg)$	0:5:50
$h(Km)$	0.5:0.2:3 (airborne application)
$V(Km)$	10:1:90, 120,200

resolution (in the order of 0.1 nm). In simulating each training sample, we randomly vary the MODTRAN parameter related to CWV according to a uniform distribution in the range $[0.5, 5]\text{g}\cdot\ \text{cm}^{-2}$. The rest of the MODTRAN parameters are constant. We refer to 1) the parameters MODEL and IHAZE that specify the geographical and seasonal model and the aerosol model; 2) the parameters that define the acquisition geometry and, specifically, h and ϑ; and 3) the visibility V. In our work, we have trained various regressors by varying the fixed parameters according to the values reported in Table I. Note that in Table I (and in the rest of the article), we use the notation "xmin:step:xmax" to indicate the set of values between "xmin" and "xmax" taken with a step size "step."

The reflectance generator block provides the surface and adjacent reflectances. They are generated starting from the spectra in the library $\mathbf{D}$ using the linear mixing model. The tth sample $\boldsymbol{\rho}_t$ is obtained by combining N randomly selected elements of $\mathbf{D}$ ($\mathbf{d}_i, i = 1, \ldots, N$) and N randomly generated mixing coefficients a_i, $i = 1, \ldots, N$

$$\boldsymbol{\rho}_t = \sum_{i=1}^{N} a_i \mathbf{d}_i. \tag{12}$$

a_is are drawn according to the Dirichlet distribution so as to meet the sum-to-one and the non-negative constraints. N is also generated as a random integer in the range [1, 5]. The same procedure is independently applied to obtain the tth adjacent reflectance vector $\boldsymbol{\rho}_{\mathbf{a},t}$.

The reflectance spectra and the radiometric quantities are then combined according to the RTM in (1) and (2) to obtain the high spectral resolution vectors $\mathbf{L}_t^*(V, \text{CWV})$ and $\mathbf{L}_{a,t}^*(V, \text{CWV})$ that are transformed in the radiances $\tilde{L}_t(V, \text{CWV})$ and $\tilde{L}_{\mathbf{a},t}(V, \text{CWV})$ measured by the sensor of reference. For this purpose, we account for the spectral response functions (SRFs) of the N_B sensor channels and, specifically, we use a Gaussian-shaped SRF for each channel, characterized by the full-width at half-maximum (FWHM) and the channel central wavelength (λ_n).

To make the regressor robust to spectral miscalibration errors and random noise, we introduced these two sources of disturbance in the learning process. In particular, to account for the spectral miscalibration, we add an offset ε_λ to the channel central wavelengths when we transform the generated radiance vectors in the sensor domain. In this work, ε_λ has values ranging from -2FWHM to 2FWHM with a step of 0.2FWHM (20 different values). It is worth noting that in this work, we have simply modeled the spectral miscalibration errors as a shift in the spectral dimension. However, the general procedure introduced here allows the use of more complex models accounting for the dependence of the error on the channels' central wavelengths.

As to random noise, we refer to the well-established signal-dependent noise model proposed in [34] and [35]. Accordingly, the noise block generates additive noise realizations as Gaussian distributed random vectors with zero mean and diagonal covariance matrix whose nth diagonal element $\sigma_{n,t}^2$ linearly depends on the at-sensor radiance in the nth channel. Namely,

$$\sigma_{n,t}^2 = \sigma_{\text{ph},n} \tilde{L}_t(\lambda_n, V, \text{CWV}) + \sigma_{\text{th},n}^2 \tag{13}$$

where $\tilde{L}_t(\lambda_n, V, \text{CWV})$ is the nth element of the radiance vector $\tilde{L}_t(V, \text{CWV})$. $\sigma_{\text{ph},n}$ and $\sigma_{\text{th},n}^2$ account for the photon noise and the thermal noise, respectively. It is assumed that the photon noise and the thermal noise have the same statistical power per band, that is, $\sigma_{\text{ph},n} \tilde{L}_t(\lambda_n, V, \text{CWV}) = \sigma_{\text{th},n}^2$. $\sigma_{\text{ph},n}$ is modeled according to [34] as

$$\sigma_{\text{ph},n} = \frac{\delta}{\lambda_n \cdot m(\lambda_n)} \tag{14}$$

where $m(\lambda_n)$ is the sensor quantum efficiency, and δ is a multiplicative factor that allows the desired signal-to-noise ratio (SNR) to be set. SNR is defined as the ratio between the total at-sensor radiance power $\sum_{n=1}^{N_b} \tilde{L}_t^2(\lambda_n, V, \text{CWV})$ and the total noise power $\sum_{n=1}^{N_b} \sigma_n^2$ and can be written as

$$\text{SNR} = \frac{\sum_{n=1}^{N_b} \tilde{L}_t^2(\lambda_n, V, \text{CWV})}{2\delta \sum_{n=1}^{N_b} \frac{\tilde{L}_t(\lambda_n, V, \text{CWV})}{\lambda_n \cdot m(\lambda_n)}}. \tag{15}$$

Equation (15) defines the dependence between the SNR and the multiplicative factor δ. As to the sensor quantum efficiency $m(\lambda_n)$, it is approximated as a Gaussian function according to the behavior experienced in real data ([34]). In the training set generation process, SNR is drawn from a uniform distribution in the range [25 dB, 60 dB].

The outputs of the whole generation process are the noisy surface radiance and adjacent radiance vectors $\mathbf{L}_t(V, \text{CWV})$ and $\mathbf{L}_{\mathbf{a},t}(V, \text{CWV})$, and the vector $\mathbf{c}_t$ of the components of $\boldsymbol{\rho}_t$ with respect to the basis $\mathbf{U}$.

All the regressors used in this work were trained using $N_T = 2 \times 10^5$ training samples obtained by varying the vectors $\boldsymbol{\rho}_t$ and $\boldsymbol{\rho}_{\mathbf{a},t}$, CWV, SNR and by considering the 20 values of the spectral offset ε_λ. During the training phase, the regressors also "learn" the value of the coefficient β in (11) by means of K-fold cross-validation. In such a procedure, we test values of β in the range $10^{[-1,\ldots,-12]}$ and we consider $K = 10$ equal sized groups of samples taken from those generated according to the procedure described in the previous part of this section. At each iteration, the kth group of data is retained as the test set and the rest of the data (those in the remaining $K - 1$ groups) are used to train the regressor. The cross-validation process is repeated K times using each group as test set once.

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The errors and the regressors obtained over the test sets in the K iterations are averaged to obtain the output error and regressor for the considered value of β. Finally, the regressor corresponding to the value of β that provides the minimum average error is selected.

In all the performed experiments, we found that a good value for β is 10^{-6}.

V. In-Scene Visibility Estimation and Algorithm Summary

In the previous sections, we have assumed that the visibility V was known. It is a fundamental parameter for selecting the specific regressor to be used. However, the value of V is generally unknown and needs to be estimated from the analyzed image. Here, we propose a method to estimate V. Let us start by denoting ψ as a certain set of bands of the adopted HS where the water vapor absorption effects are negligible. It is assumed that for each band in ψ there is at least an image pixel with negligible ground reflected radiance ($\mathbf{L}_{\mathrm{GND}}$). It is worth noting that this hypothesis may be true in several practical situations. For instance, over land scenarios, the hypothesis is likely to be true in the visible spectral range (up to 650 nm), where we can have materials with very low reflectance (i.e., vegetation) or shadowed areas that are characterized by very low values of the ground reflected radiance. Over maritime scenarios, it is generally true in the near-infrared spectral range (from 700 nm onward) where the water has very low spectral reflectance values.

Let us denote $\mathbf{L}^{\psi}$ as the vector obtained by taking the minimum over the data set (image) in each band of ψ. According to the RTM in (1), it can be considered as an approximation of $\mathbf{L}_p(\mathbf{\Gamma}, V)$ limited, of course, to the set of bands ψ. Note that we have deliberately ignored the dependence of $\mathbf{L}_p$ on CWV because the set ψ does not include water absorption spectral windows. Furthermore, we reintroduced the vector of known RTM parameters $\mathbf{\Gamma}$,to specify that $\mathbf{L}_p$ refers to the specific acquisition conditions pertaining the analyzed image.

We can use the above-mentioned approximation in conjunction with a set of $\mathbf{L}_p(\mathbf{\Gamma}, V)$ vectors simulated by MODTRAN for different values of V, to obtain the estimate $\hat{V}(\mathbf{\Gamma})$ of the visibility. Specifically, we propose to solve the following minimization problem:

$$\hat{V}(\mathbf{\Gamma}) = \arg\min_{V}\left\{E\left[\mathbf{L}^{\psi}, \mathbf{L}_p^{\psi}(\mathbf{\Gamma}, V)\right]\right\}$$

$$E\left[\mathbf{L}^{\psi}, \mathbf{L}_p^{\psi}(\mathbf{\Gamma}, V)\right] = \frac{\left\|\mathbf{L}^{\psi} - \mathbf{L}_p^{\psi}(\mathbf{\Gamma}, V)\right\|}{\left\|\mathbf{L}_p^{\psi}(\mathbf{\Gamma}, V)\right\|} + \frac{1}{N_{\psi}} u\left(\mathbf{L}_p^{\psi}(\mathbf{\Gamma}, V) - \mathbf{L}^{\psi}\right) \quad (16)$$

where N_{ψ} is the number of bands in ψ, $\mathbf{L}_p^{\psi}(\mathbf{\Gamma}, V)$ is the upwelling atmospheric vector over the set of bands ψ, and u is the Heaviside step function. The first term in the objective function E accounts for the relative Euclidean distance between $\mathbf{L}^{\psi}$ and $\mathbf{L}_p^{\psi}(\mathbf{\Gamma}, V)$. The second term is introduced to consider that according to the physical model in (1), $\mathbf{L}^{\psi} \geq \mathbf{L}_p^{\psi}(\mathbf{\Gamma}, V)$. The problem in (16) is numerically solved, calculating the objective function over the vectors $\mathbf{L}_p^{\psi}(\mathbf{\Gamma}, V)$ derived by MODTRAN on a discrete set of values for V.

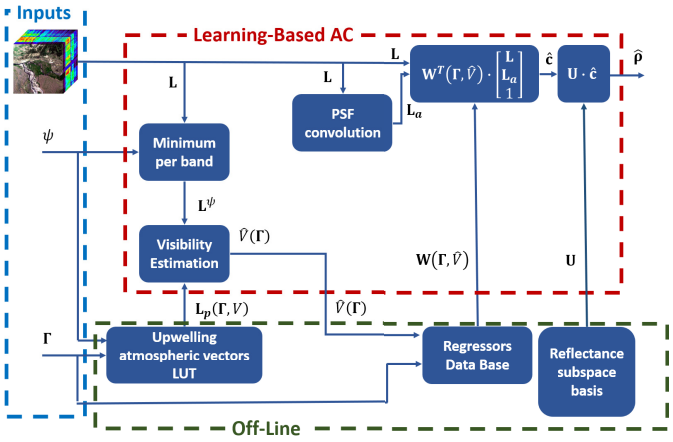


Fig. 2. Block diagram of the LBAC algorithm.

Note that in practice, to deal with noise issues, $\mathbf{L}^{\psi}$ can be obtained by averaging the P (typically 10) smallest values of the data over each band.

The entire learning-based AC (LBAC) procedure is summarized in the block diagram of Fig. 2. It receives as inputs the hyperspectral radiance image, the known parameters $\mathbf{\Gamma}$ (MODEL, IHAZE, h, ϑ) defining the acquisition conditions, and the set of bands ψ for the visibility estimation. It derives the vectors $\mathbf{L}^{\psi}$ that are compared with the values of $\mathbf{L}_p(\mathbf{\Gamma}, V)$ recorded in the "upwelling atmospheric LUT" to solve the minimization problem in (16), providing as output the visibility estimate $\hat{V}(\mathbf{\Gamma})$. This estimate along with the vector $\mathbf{\Gamma}$ allows the regressor $\mathbf{W}(\mathbf{\Gamma}, \hat{V})$ to be selected from the "Regressors Database." Specifically, the regressor learned with the value of V closer to $\hat{V}(\mathbf{\Gamma})$ is selected. Furthermore, the image is convolved with the PSF kernel to obtain the adjacent radiance vector $\mathbf{L_a}$. Finally, the regressor is applied to the vectors $\mathbf{L}$ and $\mathbf{L_a}$ to obtain the reflectance vector estimate $\hat{\rho}$.

It is worth noting that the "upwelling atmospheric LUT" is derived off-line (and not during the algorithm application) by means of MODTRAN. The "Regressors Database" is also derived off-line by the learning strategy described in Section IV.

The core of the algorithm is the regressor application whose computational cost can be easily quantified. In fact, for each image pixel it consists in the matrix multiplication between the $(2N_B + 1) \times K$ regressor matrix $\mathbf{W}(\mathbf{\Gamma}, \hat{V})$ and the $(2N_B + 1) \times 1$ vector $\mathbf{x} = [\mathbf{L}^{\mathrm{T}}, \mathbf{L_a}^{\mathrm{T}}, 1]^{\mathrm{T}}$, which has a computational complexity of $O((2N_B + 1) \cdot K)$. To obtain the reflectance estimate, the $K \times 1$ coefficients' vector $\hat{c}$ is multiplied for the $N_B \times K$ basis matrix $\mathbf{U}$ with a computational complexity of $O(N_B \cdot K)$. The total computational complexity for the reflectance estimation is $O((3N_B + 1) \cdot K)$.

VI. Results on Simulated Data

In this section, we test the LBAC over simulated data to evaluate its performance and to show the robustness of the

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algorithm to both the random noise and spectral miscalibration errors. We would like to remark that the robustness to noise and spectral errors is obtained by including these two sources of disturbance in the learning strategy. To show the effectiveness of that learning strategy, the results provided by LBAC will be compared with those obtained by the same algorithm when the training phase is performed without including the random noise or the spectral miscalibration errors in the generation of the training set.

The test data are generated according to the simulation procedure described in Section IV. Specifically, we start by generating $N = 10^5$ couples of reflectance vectors ($\boldsymbol{\rho}$) and adjacent vectors ($\boldsymbol{\rho}_\mathbf{a}$). These vectors are combined with the MODTRAN-generated radiometric quantities to obtain the sets of the surface radiance vectors ($\mathbf{L}$) and adjacent radiance vectors ($\mathbf{L}_\mathbf{a}$), which represent the simulated image. For each simulated image, we set the MODTRAN parameter MODEL to 2 (Mid-Latitude Summer) and we randomly generate a CWV value for each pixel so as to simulate the spatial variability of that parameter. Specifically, the CWV values are obtained as realizations of a uniformly distributed random variable in the range $[0.5, 5]\mathrm{g}\cdot f\,\mathrm{cm}^{-2}$. The rest of the MODTRAN parameters are kept constant over the image pixels and are generated randomly. Specifically, the values of IHAZE are selected randomly among those in Table I; ϑ is drawn from a uniform distribution in the range $[0, 50]°$; assuming an aerial scenario, h is generated as uniformly distributed in the range $[0.5, 3]$ km. The visibility values are randomly generated in the range $[10, 120]$ km.

Using this procedure, we simulated 100 different radiance images (including both surface radiance vector and adjacent radiance vectors) with $N = 10^5$ pixels resulting in a total number of test samples equal to 10^7. Note that the procedure adopted to simulate the test set is the same as described in Section IV, but, here, we use new randomly generated CWV values and new randomly generated reflectance spectra [according to (12)]. Thus, the generated test set is unlikely to contain the same samples of the training set.

As to the HS model, we refer to the SIM.GA ("Sistema Iperspettrale Modulare. Galileo Avionica," [36]) sensor. It is an avionic hyperspectral instrument manufactured by Leonardo-Finmeccanica that operates in the VNIR spectral range (from 400 to 1000 nm). SIM.GA has a spectral sampling of 1.2 nm and FWHM = 2.284 nm resulting in 512 spectral channels. Generally, data are preprocessed by binning adjacent channels with a factor of 2 so as to obtain 256 spectral bands.

To test the robustness to the noise sources, we considered several cases with different SNRs and spectral error offsets (ε_λ) that will be detailed in the discussion of each performed experiment.

As to LBAC, the reflectance subspace matrix $\mathbf{U}$ is determined by the procedure in Appendix I and using the spectral dictionary $\mathbf{D}$ presented in Section IV. The adopted matrix $\mathbf{U}$ has rank $K = 25$. The set of bands ψ adopted for visibility estimation refers to the VIS spectral range and contains the bands corresponding to the central wavelengths included in the range [400–650] nm (114 bands).

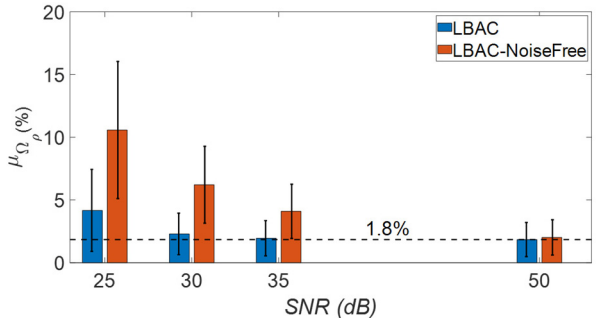


Fig. 3. Values of μ_{Ω_ρ} (in percent) obtained by LBAC and LBAC-NoiseFree for various SNRs (25, 30, 35, and 50 dB). In this figure, we also show the std of Ω_ρ in the form of error band.

According to the premise of this section, we consider three different versions of the algorithm that will be referred to as 1) LBAC that is trained by accounting for both the random noise and spectral errors; 2) LBAC-NoiseFree that is trained without taking into account the random noise and spectral miscalibration errors; and 3) LBAC-MirRegFree that is trained without taking the spectral miscalibration errors into accounts ($\varepsilon_\lambda = 0$), but accounting for random noise. The number of training samples is the same for all the above-mentioned versions of LBAC.

To assess the performance of the algorithms, we refer to the relative percentage error between the estimate $\hat{\rho}$ and the real value $\boldsymbol{\rho}$, that is, $\Omega_\rho = (\|\boldsymbol{\rho} - \hat{\rho}\|_2 / \|\boldsymbol{\rho}\|_2) \times 100$. It a random variable that is characterized through the expectation $\mu_{\Omega_\rho} = E\{\Omega_\rho\}$ and the probability of exceedance (PoE) defined as the complement of the cumulative distribution function (cdf)

$$\mathrm{PoE}(\Omega_\rho) = 1 - \mathrm{cdf}(\Omega_\rho). \tag{17}$$

In the first experiment, we assess the performance on test sets affected by random noise and assumed to be correctly calibrated in the spectral dimension (miscalibration errors absent). Specifically, we consider four different cases: 1) SNR = 50 dB (very low noise); 2) SNR = 35 dB (moderate noise); 3) SNR = 30 dB (high noise); and 4) SNR = 25 dB (very high noise). To show the effectiveness of the learning strategy accounting for random noise, we compare the performance of LBAC with that of LBAC-NoiseFree. In Fig. 3, we show the values of μ_{Ω_ρ} obtained in the four SNR cases. For completeness, we also show the standard deviation (std) of Ω_ρ in the form of error bar. We can note that with very low noise (SNR = 50dB), both the algorithms have high accuracy with a mean error of about 1.8%. Decreasing the SNR the performance worsens. However, the LBAC provides values of the mean error much smaller than that of the LBAC-NoiseFree. For instance, in the case of very high noise (SNR = 25 dB), the LBAC has $\mu_{\Omega_\rho} = 4\%$, while the LBAC-NoiseFree provides a mean error value greater that 10%.

The analysis of the entire distribution of the error carried out in terms of PoE confirms the previous conclusions. Fig. 4(a) and (b) shows PoE for various SNR values for LBAC and LBAC-NoiseFree, respectively. As to LBAC, we can note that except for the case of SNR = 25dB (very high

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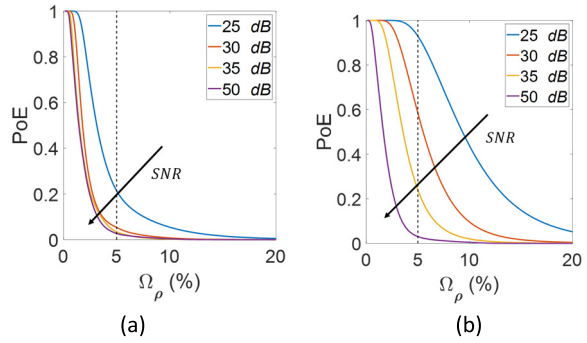


Fig. 4. PoE for the errors obtained by (a) LBAC and (b) LBAC-NoiseFree for various SNRs (25, 30, 35, and 50 dB).

TABLE II
VALUES OF PoE(5%) FOR LBAC AND LBAC-NOISEFREE FOR DIFFERENT SNRS

$SNR\ (dB)$	LBAC $PoE(5\%)$	LBAC-NoiseFree $PoE(6\%)$
25	$2.1 \cdot 10^{-1}$	$9.2 \cdot 10^{-1}$
30	$4.3 \cdot 10^{-2}$	$5.8 \cdot 10^{-1}$
35	$3.4 \cdot 10^{-2}$	$2.5 \cdot 10^{-1}$
50	$2.7 \cdot 10^{-2}$	$3.0 \cdot 10^{-2}$

noise), the PoE curves are very close to each other, thus showing the robustness of the method to the presence of noise. LBAC-NoiseFree shows a greater dependence on noise than LBAC. The PoE curves obtained for different values of SNR are quite distinct. Furthermore, with the same SNR, the PoE provided by the LBAC-NoiseFree are well above those of the LBAC, thus proving that the estimation errors for LBAC-NoiseFree are in general greater than those provided by LBAC.

To give a numerical example, Table II shows the values of PoE at $\Omega_\rho = 5\%$ for the two algorithms and the considered SNRs. Specifically, the values in Table II indicate, for each considered case, the probability of having an estimation error Ω_ρ greater than 5%. We note that the values of PoE(5%) provided by LBAC are approximately 1 order of magnitude smaller than those provided by LBAC-NoiseFree for each of the considered SNR case. Of course, the algorithms attain the same value in the case of very low noise (SNR = 50 dB). Table II also confirms that for SNR = 30 dB, SNR = 35 dB, and SNR = 50 dB, the LBAC provides approximately the same results.

In Fig. 5, we show an example of the results provided by the algorithms. Fig. 5(a) and (b) refers to the case of SNR = 30 dB. In particular, Fig. 5(a) shows the surface spectral radiance and the adjacent spectral radiance provided as input to the regressor, while Fig. 5(b) shows the comparison of the reflectance estimates provided by LBAC and LBAC-NoiseFree with the true reflectance spectrum adopted to generate the input data of Fig. 5(a). Note that in Fig. 5(a), we show both the original simulated radiances (dashed lines) and their noisy versions (the inputs of the regressors) to give an idea of the disturbance introduced by the random noise with SNR = 30dB. The results in Fig. 5(b) give an example of the best performance provided in the presence of noise by LBAC with respect to LBAC-NoiseFree (trained without accounting for the random noise). In fact, the reflectance estimate obtained by LBAC is very close to the true one, while that provided by LBAC-NoiseFree exhibits several artifacts that are due to the noise affecting the input data. For completeness, in Fig. 5(c), we report the results provided by the two regressors in the case of very low noise (SNR = 50dB). As expected, in the presence of low noise the two algorithms provide approximately the same (good) result. The transparent blue patches in Fig. 5(a)–(c) highlight the water absorption bands.

The second experiment is designed to test the proposed procedure in the presence of both the spectral misregistration errors and random noise. For this purpose, we set SNR = 35dB and $\varepsilon_\lambda = t$FWHM with $t = [-2 : 0.5 : 2]$. In this case, we compare LBAC with the version of the regressor trained without accounting for the spectral miscalibration errors (LBAC-MisRegFree). Fig. 6 shows the values of μ_{Ω_ρ} (and the std of Ω_ρ) in all the cases tested. We can note that the estimation performance worsens with the increase in the absolute value of the spectral offset ε_λ. However, the LBAC provides the lowest value of μ_{Ω_ρ} for each considered spectral offset. This proves the effectiveness of the proposed learning strategy in making LBAC less sensitive to the spectral misregistration errors. With ε_λ ranging from -1.5FWHM to 1.5FWHM, the LBAC has high estimation accuracy with a mean error lower than 3%.

Analysis of PoE confirms the previous conclusions. In Fig. 7(a) and (b), we show PoE for LBAC and LBAC-MisRegFree obtained with $t = [-2, 0, 2]$. We can see that the two algorithms attain approximately the same performance in the absence of misregistration errors. In the case of LBAC, the PoE curves with $t = [-2, 2]$ are closer to the PoE in the absence of spectral miscalibration errors than in the case of LBAC-MisRegFree. Furthermore, the curves obtained by LBAC-MisRegFree are above those provided by LBAC for both $t = -2$ and $t = 2$.

TABLE III
POE(5%) FOR LBAC AND LBAC-MISREGFREE FOR DIFFERENT VALUES OF t AND SNR = 35 DB

t	LBAC $PoE(5\%)$	LBAC-MisRegFree $PoE(6\%)$
-2	$1 \cdot 10^{-1}$	$3.1 \cdot 10^{-1}$
0	$2.3 \cdot 10^{-2}$	$2.4 \cdot 10^{-2}$
2	$3 \cdot 10^{-1}$	$8.4 \cdot 10^{-1}$

To give an example in Table III, we show for both the algorithms the values of PoE at $\Omega_\rho = 5\%$ for $t = [-2, 0, 2]$. Note that as expected, in the absence of spectral misregistration errors, the algorithm provides the same performance: the probability that the estimation error is greater than 5% is 2.3×10^{-2} for LBAC and 2.4×10^{-2} for LBAC-MisRegFree.

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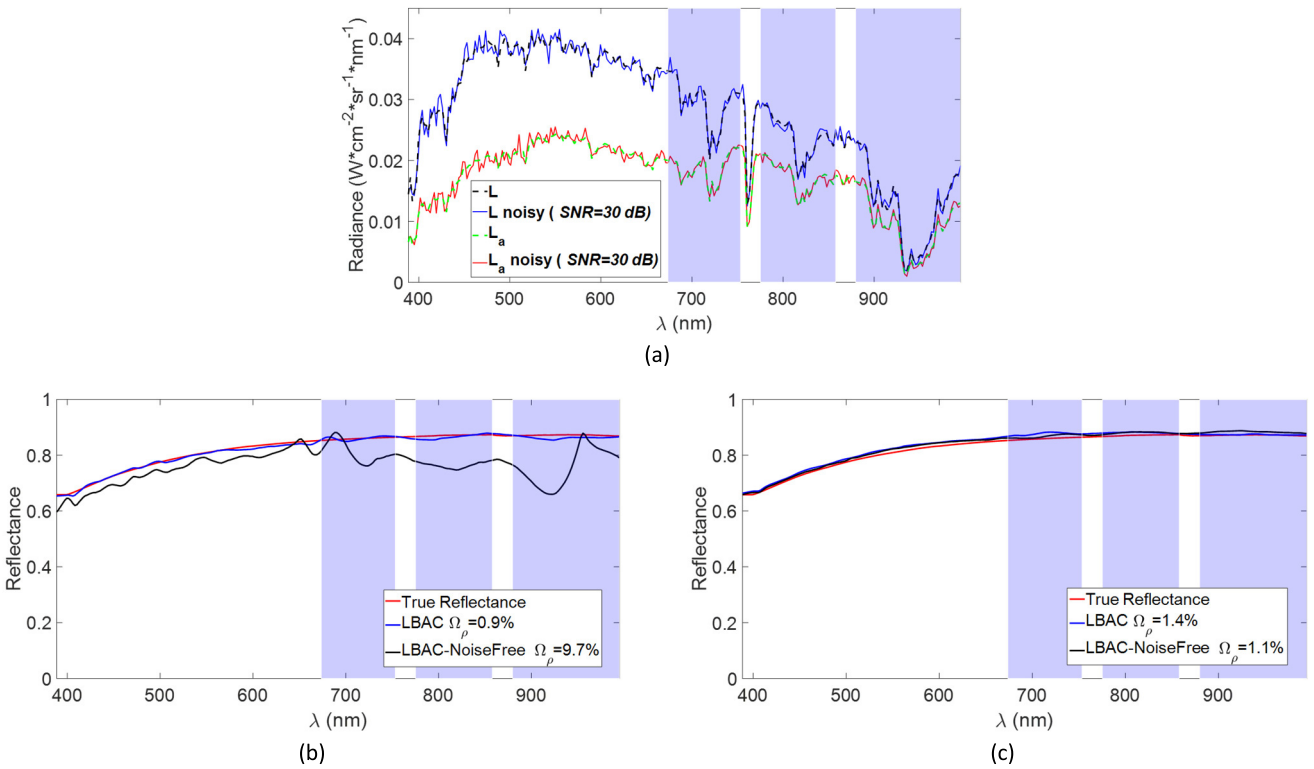


Fig. 5. (a) Surface spectral radiance (noise-free and noisy version with SNR = 30 dB) and adjacent spectral radiance (noise-free and noisy version with SNR = 30 dB) provided as input to the regressor. (b) Comparison of the estimation results provided by LBAC and LBAC-NoiseFree (in the case of SNR = 30 dB) with the real reflectance spectrum. (c) Comparison of the estimation results provided by LBAC and LBAC-NoiseFree (in the case of SNR = 50 dB) with the real reflectance spectrum.

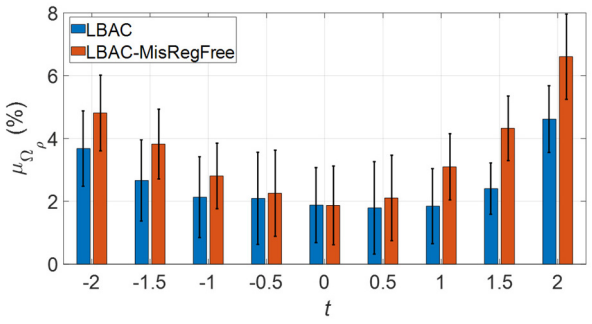


Fig. 6. Values of μ_{Ω_ρ} (in percent) obtained by LBAC and LBAC-MisRegFree for $\varepsilon_\lambda = t\text{FWHM}$ ($t \epsilon [-2 : 0.5 : 2]$) and SNR = 35 dB. In this figure, we also show the std of Ω_ρ in the form of error band.

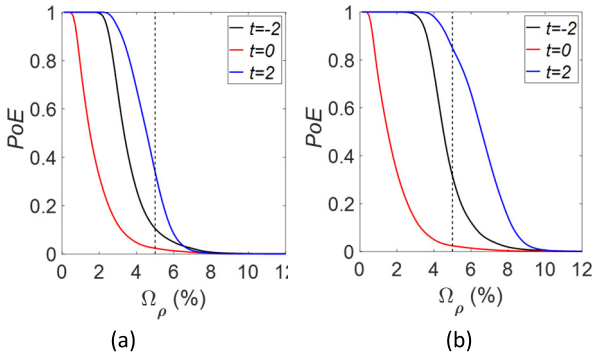


Fig. 7. PoE for (a) LBAC and (b) LBAC-MisRegFree obtained with $\varepsilon_\lambda = t\text{FWHM}$ and $t = [-2, 0, 2]$ and SNR = 35 dB.

In the other two cases, LBAC has PoE values lower than those attained by LBAC-MisRegFree.

In Fig. 8, we show two examples of the results provided by LBAC and LBAC-MisRegFree in the case of $t = -1$ [Fig. 8(a)] and $t = 1$ [Fig. 8(b)] and with SNR = 35dB. From Fig. 8(a) and (b), we compare the true reflectance spectra with those estimated by the two algorithms. We can note that the miscalibration errors induce artifacts in the estimated spectra. However, the artifacts are less pronounced in the estimates provided by LBAC.

It is worth noting that one of the parameters to be set in LBAC is the rank K of the spectral reflectance subspace. Such parameters have been set to 25 in all the presented experiments. Of course, the value of K impacts on the performance of the algorithm. Specifically, for small values of K, the LBAC cannot accurately reconstruct the reflectance spectrum from the observed radiance. To give an example, in the following we show the results obtained on a test set simulated by setting MODEL = 2, IHAZE = 1, h= 2km, and V= 20km. CWV is generated as uniformly distributed in the range [0.5, 5]g·cm^{-2}, and the noise parameters SNR and t have been set to 35dB and 0 (no spectral misregistration), respectively.

Regressors trained by considering different values for the parameter K ($K = [2 : 2 : 50]$) have been applied to the

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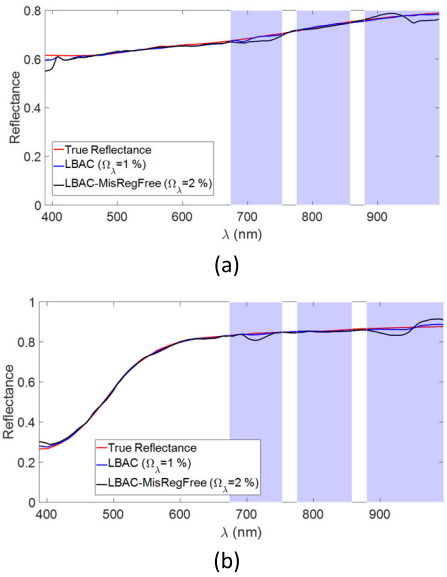


Fig. 8. Spectral reflectances estimated by LBAC and LBAC-MisregFree in the two cases of (a) $t = -1$ ($\varepsilon_\lambda = -\text{FWHM}$) and (b) $t = 1$ ($\varepsilon_\lambda = \text{FWHM}$). In both the cases, SNR is set to 35 dB.

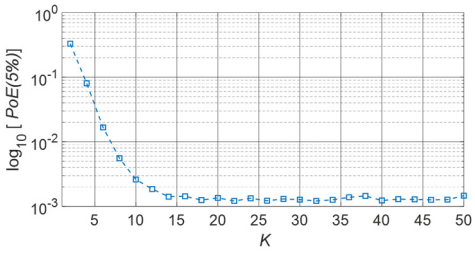


Fig. 9. Values of the PoE as a function of K.

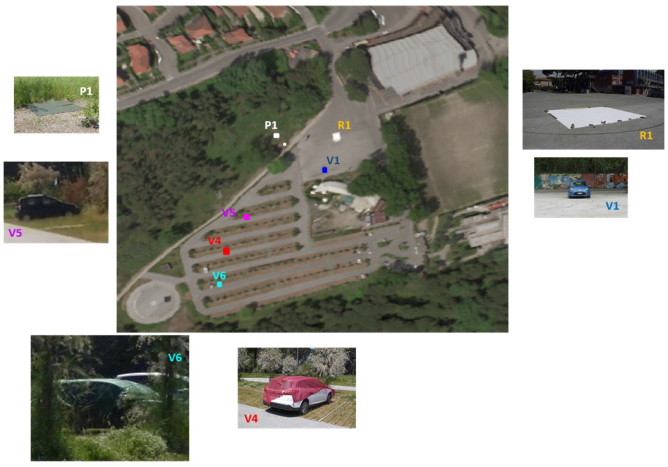


Fig. 10. RGB representation of D2F22H2 image. This figure shows the position of the targets and a photograph of each of them.

above-mentioned data set. In Fig. 9, we show the probability of having an error Ω_ρ greater than 5% [PoE(5%)] for all the applied regressors.

Note that for K ranging from 2 to 20, LBAC performance increases by decreasing K. The values of PoE(5%) are approximately constant for $K > 20$. It is important to remark that the



Fig. 11. RGB representation of D2F21H2 image. This figure shows the position of the targets and a photograph of each of them.

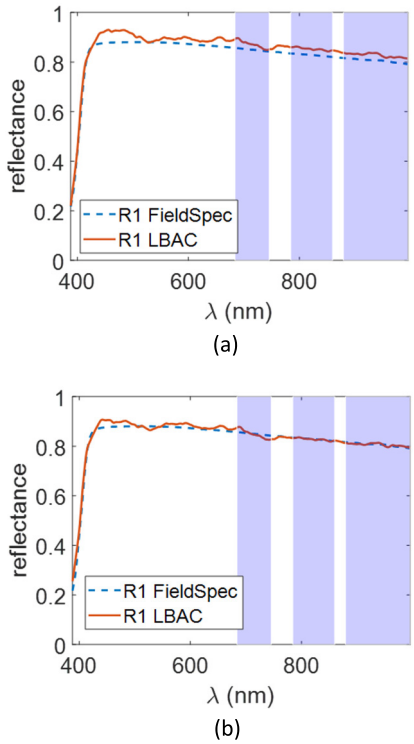


Fig. 12. (a) D2F22H2, Target R1: comparison between LBAC reflectance estimate and the FieldSpec measured spectrum. (b) D2F21H2, Target R1: comparison between LBAC reflectance estimate and the FieldSpec measured spectrum.

computational load of the algorithm increases linearly with K. Thus, K should be chosen as small as possible.

VII. Results Over Real Data

To conclude the experimental part of this article, we show the results obtained by applying LBAC to real images acquired by two airborne HSs. The images are part of two data sets named "Viareggio 2013 Trial" ([36]) and "SHARE 2012" ([37]). Specifically, we consider two images named D2F22H2 and D2F21H2 from the first data set and one image named AVON-morning from the second data set.

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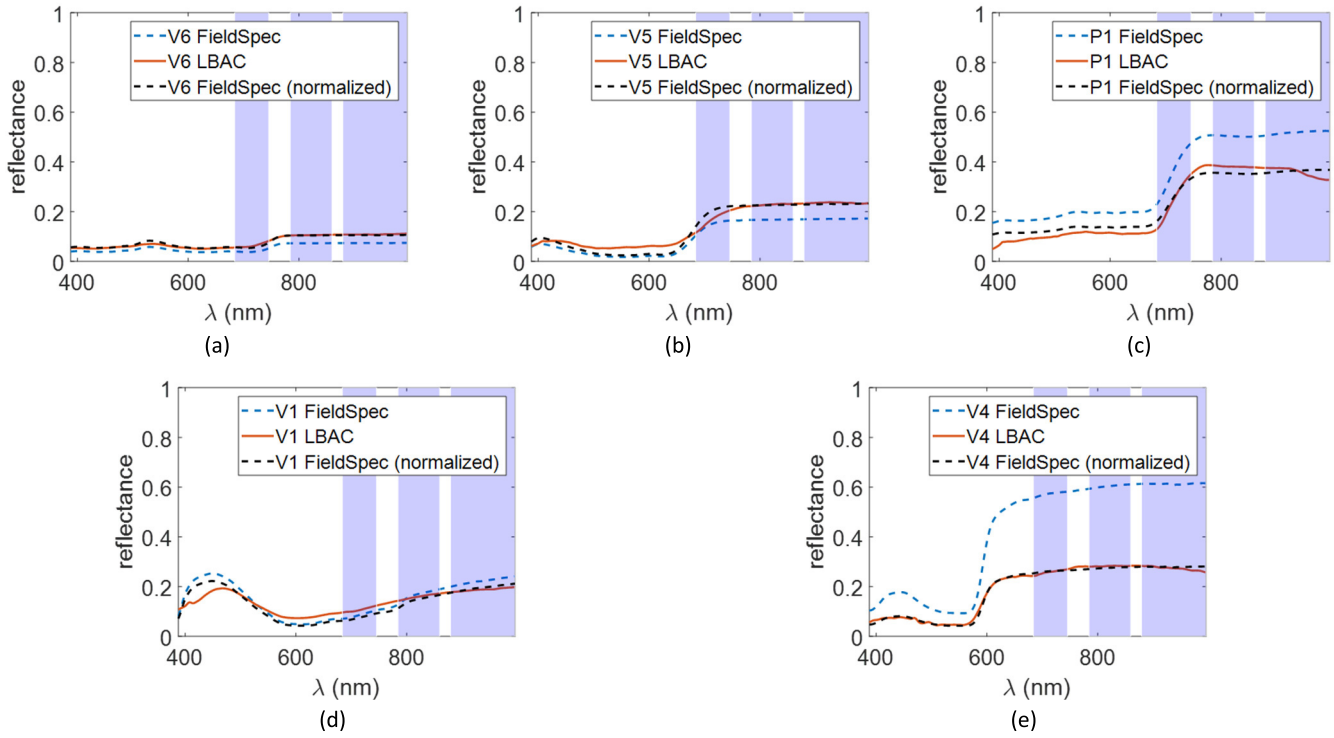


Fig. 13. D2F22H2: comparison between the reflectance spectrum measured by the FieldSpec instrument, its scaled version, and that obtained from the image by LBAC for the ground truth targets: (a) V6, (b) V5, (c) P1, (d) V1, and (e) V4.

A. Results on D2F22H2 and D2F21H2

D2F22H2 and D2F21H2 were collected by the SIM.GA sensor introduced in Section IV on May 9, 2013, at 12:38 and 9:49 [Greenwich Mean Time (GMT)], respectively. Both the images were acquired under clear sky conditions over the town of Viareggio (Italy) and concern a scene including a parking lot (where a number of man-made objects were deployed), several sport facilities buildings, and a football field. The flight height in both the acquisitions was about 1 km corresponding to a nominal ground spatial resolution of about 0.6 m. The images are equipped with the ground truth data including the map of the deployed man-made objects (targets) and their spectral reflectances measured by an ASD FieldSpec hand-held spectroradiometer.

The red green blue (RGB) representations of the two images are shown in Figs. 10 and 11. In each figure, the locations of the man-made objects are highlighted.

The regressors adopted in this experiment were chosen among those trained by setting the MODTRAN parameters MODEL to 2 (Mid-Latitude summer), IHAZE to 4 (Maritime aerosol model), h according to the flight altitude, and ϑ according to the coordinates of the scene and the acquisition date and time. The values of MODEL and IHAZE are motivated by the geographical coordinates of the scene (in close proximity of the Tyrrenian Sea) and by the fact that acquisition was made in the late spring season. The specific regressor adopted in the two images is selected on the basis of the V estimates provided by the visibility estimation procedure described in Section V, that is, $\hat{V} = 31$km for D2F22H2 and $\hat{V} = 34$km for D2F21H2. In applying the visibility estimation procedure, we considered the set of bands ψ in the VIS spectral range corresponding to the bands with central wavelengths from 400 to 650 nm (114 bands). The adopted matrix $\mathbf{U}$ has rank $K = 25$ and is determined by the procedure described in Appendix I.

The adjacent radiance spectra are obtained by convolving the images with a Gaussian shaped kernel (simulating the atmospheric PSF) with std equal to 2 pixels.

The target labeled as R1 in both the images is a white calibration tarp that was adopted for its isotropic reflective behavior that makes it a good approximation of a Lambertian surface. In Fig. 12(a) and (b), we show the reflectance estimates (solid line) provided by LBAC in the pixel corresponding to R1 in D2F22H2 and D2F21H2, respectively. Fig. 12(a) and (b) also shows the spectral reflectance of R1 measured by the FieldSpec instrument (dashed line) and included in the ground truth metadata. The transparent blue patches highlight the spectral windows corresponding to water vapor absorption.

The high similarity among the spectral reflectance estimates provided by LBAC on the two images and the measured spectrum show the effectiveness of the proposed technique. It is worth noting that the water absorption effects are correctly compensated.

We also compare the LBAC estimates with the measured spectra for all the targets in the ground truth of the two images. Specifically, in Fig. 13 we show the results obtained on the D2F22H2 image, while in Fig. 14 we show the results obtained on the D2F21H2 image. It is worth noting that the RTM adopted to define LBAC does not account for the

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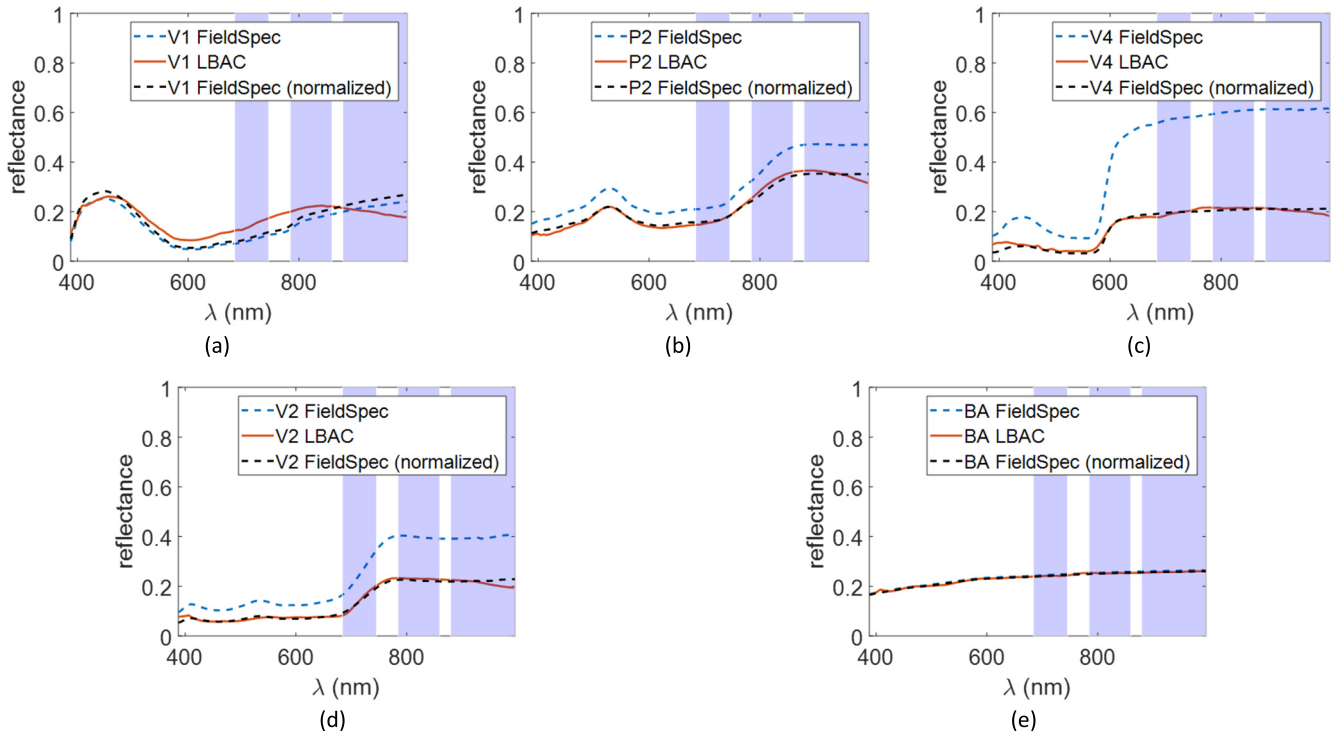


Fig. 14. D2F21H2: comparison between the reflectance spectrum measured by the FieldSpec instrument, its scaled version, and that obtained from the image by LBAC for the ground truth targets: (a) V1, (b) P2, (c) V4, (d) V2, and (e) BA.

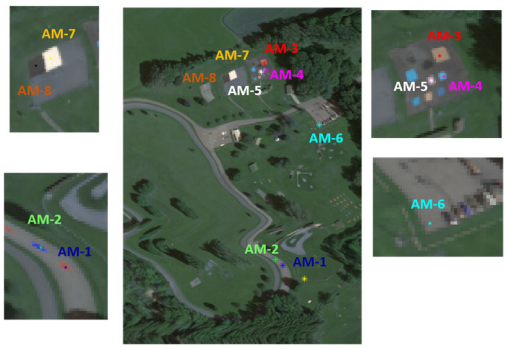


Fig. 15. RGB representation of the AVON-morning image. This figure shows the position of the targets.

bidirectional reflectance distribution function whose effect is generally roughly approximated as a scaling factor ([6], [38]). For this reason, in Figs. 13 and 14, we also plot the versions of the measured FieldSpec spectra scaled so as to have the same energy of the LBAC retrieved spectra. In Figs. 13 and 14, we can note that LBAC provides very good results.

B. Results on AVON-Morning

AVON-morning image was acquired on September 20, 2012, at 15:46 (GMT), in a rural scene located in AVON, New York, just south of Rochester. It was collected by ProSpecTIR VS system, a hyperspectral airborne pushbroom spectrometer operating in the VNIR and shortwave infrared (SWIR) spectral ranges. In this work, we process only channels 1–144 covering the spectral region from 400 to 1100 nm (VNIR). The radiance image is available on line at https://www.rit.edu/cos/share2012/. The reflectance image obtained by applying the commercial software ATCOR is also available [8]. ATCOR reflectance data are used in this experiment to validate the results provided by LBAC. Specifically, eight pixels are considered corresponding to different objects/materials, that is, tarps AM-3, AM-4, and AM-5, calibration panels AM-7 (white) and AM-8 (black), asphalt AM-6, and vehicles AM-1 and AM-2.

In Fig. 15, we show the RGB representation of the image along with the selected targets.

As to LBAC, the regressors adopted in this experiment were chosen among those trained by setting the MODTRAN parameters MODEL to 2 (Mid-Latitude summer), IHAZE to 1 (Rural aerosol model), h to 0.92 (according to the flight data), and ϑ according to the coordinates of the scene and the acquisition date and time. Here again, the values of MODEL and IHAZE are motivated by the geographical coordinates of the scene and by the fact that acquisition was made in the late summer season. According to the proposed algorithm, the specific regressor adopted is selected on the basis of the V estimate obtained by applying the visibility estimation procedure, that is, $\hat{V} = 190$km. For this purpose, we considered ψ as the first 55 channels falling in the VIS spectral range (from 400 to 650 nm). Similar to the experiment discussed in Section VII-A, the matrix $\mathbf{U}$ has rank $K = 25$.

The results of LBAC (solid lines) are compared with those of ATCOR (dashed lines) in Fig. 16, while in Table IV we show the relative error $\boldsymbol{\Omega}_\rho$ (defined in Section VI) between each LBAC retrieved spectrum and that provided by ATCOR.

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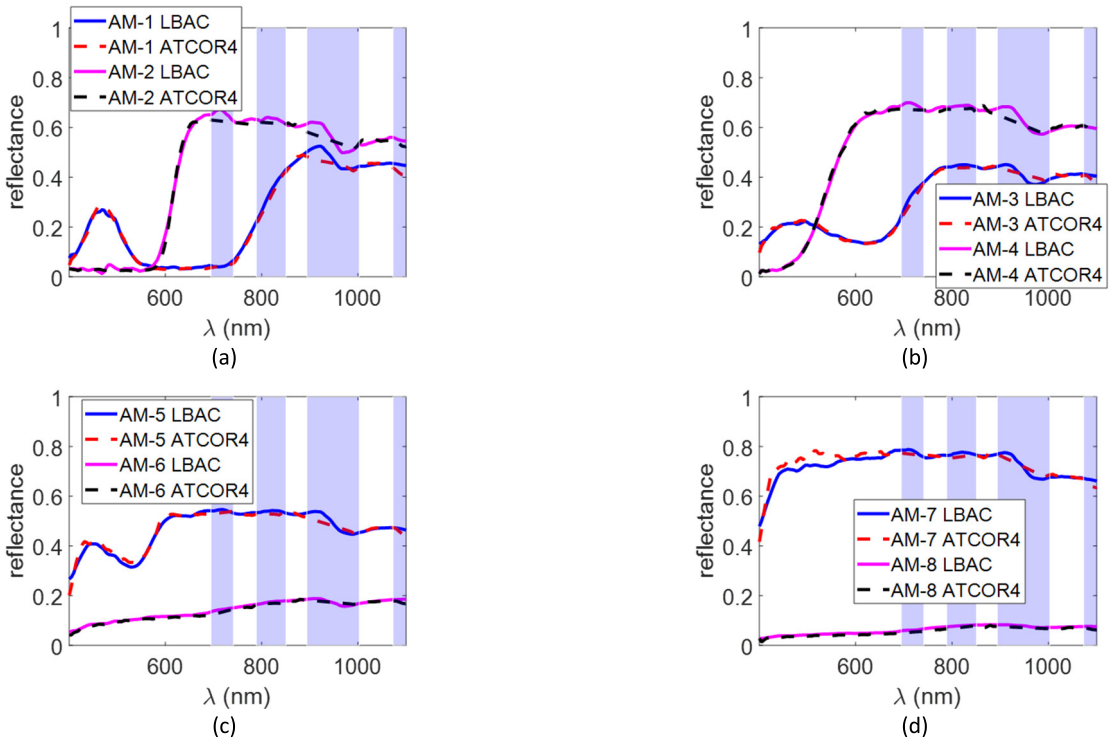


Fig. 16. AVON-morning: comparison among the spectral reflectance provided by LBAC (solid line) and those obtained by ATCOR (dashed lines) for the targets: (a) AM-1, AM-2; (b) AM-3, AM-4; (c) AM-5, AM-6; (d) AM-7, AM-8.

TABLE IV
VALUES OF THE RELATIVE ERRORS BETWEEN LBAC ESTIMATES AND ATCOR RETRIEVED SPECTRA

Target Name	LBAC Ω_ρ (%)
AM-1	4.9
AM-2	4.4
AM-3	3.4
AM-4	2.7
AM-5	3.1
AM-6	5.1
AM-7	3.1
AM-8	4.1

Fig. 16(a)–(d) shows that for all the considered objects/materials, the spectra derived by LBAC are very similar to that obtained by ATCOR. This conclusion is confirmed by the relative error values of Table IV, which are all less than 5%.

For completeness, we have computed the relative error Ω_ρ all over the image, obtaining a mean relative error of about 4.4% that confirms the closeness between the results provided by LBAC and ATCOR.

VIII. CONCLUSION

In this work, the LB approach for AC of VNIR hyperspectral data has been introduced. It aims at estimating the reflectance vectors from the observed at-radiance image using a parametric regressor (g) whose parameters are determined through a specific learning strategy based on synthetic data. The training set is obtained using a simulation procedure that accounts for the radiative transfer in the atmosphere by means of MODTRAN-generated radiometric quantities and for the variability of the surface spectral reflectance by exploiting the available spectral libraries. The learning strategy is designed to account for the effects of signal-dependent random noise and spectral miscalibration errors to teach the regressor to compensate for these effects.

According to the general framework of the LB approach, we have proposed a specific regressor based on a multilinear model that starting from the knowledge of the atmospheric visibility compensates the water absorption and provides the spectral reflectance of each pixel of the analyzed image. The visibility is derived by the image itself by means of a specific procedure exploiting image pixels characterized by low ground-leaving radiance.

The experiment over simulated data has shown the effectiveness of the proposed method. Specifically, the results on simulated data have shown the effectiveness of the learning strategy which involves the noise sources to train a regressor capable of compensating the effects of signal-dependent random noise and miscalibration errors.

Experiments on three real hyperspectral images have been presented. The obtained results have confirmed the effectiveness of the proposed LBAC that provided spectral reflectances very close to the ground truth data in the case of D2F22H2 and D2F21H2 images and very similar to those obtained by ATCOR in the case of AVON-morning image.

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It is worth noting that the algorithm in its present form does not account for shadow effects and is specifically conceived to work in clear sky conditions. Besides the effects of shadow and clouds, several open issues still remain such as those related to the direct estimation of the aerosol model or those related to the estimation of the atmospheric PSF. Furthermore, although LBAC is expected to provide satisfactory results on spectra having a smooth behavior (as in the case of most natural and man-made materials), it may not be suitable for preserving the narrow band spectral features present in specific spectra (as in the case of gases or minerals). The ability to deal with narrow band spectral features needs to be further investigated and represents one of the future directions of the research activity concerning the learning-based approach.

However, the introduced general LB framework could be a very useful tool for approaching more complicated problems. Indeed, by acting on both the learning strategy and the architecture of the regressor, the phenomena with increasing complexity could be faced. Accurate models for the physical phenomena of interest could be involved in the learning strategy to train regressors capable of accounting for their effects. Furthermore, using complex architectures, such as neural or deep networks with varying number of hidden layers, complicated data structures could be learned so as to achieve high precision and flexibility in solving very complex problems.

We conclude this section by providing a numerical example of the computational time required by the proposed procedure. Specifically, using a MATLAB implementation of LBAC on a personal computer equipped with Intel Core i7-4820K CPU at 3.7 GHz and with a 32 GB of RAM, we experienced a mean computing time per pixel of about 48μs in the case of D2F22H2 and D2F21H2 images (256 spectral bands and K =25) and of about 40μs in the case of AVON-morning image (175 spectral bands and K =25). Of course, the processing time can be significantly reduced using optimized software and dedicated hardware or modern graphical processing units.

APPENDIX I

In this section, we detail the procedure for the estimation of the reflectance subspace basis matrix $\mathbf{U}$ from the dictionary $\mathbf{D}$. This latter can be obtained using the numerous reflectance spectral library made available for the scientific community. In this work, we consider the dictionary obtained by merging three spectral libraries named ASTER ([31]), USGS (version released in 2007, [32]), and ANGERS ([33]). They contain different reflectance spectra referring to man-made objects, soil materials, minerals, vegetation, and water, for a total of 1627 samples.

Several strategies can be adopted to learn $\mathbf{U}$ from the dictionary$\mathbf{D}$. Here, we exploit the idea that the variability of each atom $\mathbf{d}_k$ of $\mathbf{D}$ around its mean value can be represented as a linear combination of a low number of vectors. Specifically, let us define $\tilde{d}_k = \mathbf{d}_k - \mu_{d_k}$ as the vector obtained by removing the average value along the spectral dimension from the kth atom of $\mathbf{D}$ and $\tilde{D}$ as the new dictionary made up of the atoms $\tilde{d}_k$. We search for the lowest rank basis matrix $\tilde{U}$ that solves the following constrained optimization problem:

$$\tilde{U} = \arg\min_{\bar{\mathbf{U}}} \sum_{k=1}^{N_D} \left\| \tilde{d}_k - \mathbf{P}_{\bar{U}} \cdot \tilde{d}_k \right\|^2$$
$$\text{s.t.}: \sum_{k=1}^{N_D} \left\| \tilde{d}_k - \mathbf{P}_{\bar{U}} \cdot \tilde{d}_k \right\|^2 < \tau \sum_{k=1}^{N_D} \left\| \tilde{d}_k \right\|^2 \quad \tau \in]0, 1] \tag{I-1}$$

where $\mathbf{P}_{\bar{U}}$ is the projection matrix for the subspace spanned by $\bar{U}$ and is defined as $\mathbf{P}_{\bar{U}} = \bar{U} \cdot (\bar{U}^T \cdot \bar{U})^{-1} \cdot \bar{U}^T$, and N_D is the number of atoms in $\tilde{D}$.

The solution of the problem in (I-1) is obtained by exploiting the SVD ([39]) and specifically by retaining the $\tilde{K}$ singular vectors of the dictionary $\tilde{D}$ corresponding to the highest singular values such as

$$\frac{\sum_{k=1}^{\tilde{K}} \sigma_k^2}{\sum_{k=1}^{N_d} \sigma_k^2} < 1 - \tau \tag{I-2}$$

where σ_k^2 is the kth singular value of $\tilde{D}$ and $\sigma_1^2 \geq \sigma_2^2 \geq \ldots \geq \sigma_{N_D}^2$. Of course, to have a good subspace representation, τ should be close to 0. In this work, we set $\tau = 10^{-4}$, thus obtaining $\tilde{K} = 24$.

To obtain the basis matrix $\mathbf{U}$ for the dictionary $\mathbf{D}$ (and then for the reflectance subspace), the unit vector ($\mathbf{1}$) needs to be added to $\tilde{U}$ to account for the average spectral value of each vector

$$\mathbf{U} = \left[\tilde{U}, \mathbf{1}\right]. \tag{I-3}$$

The resulting basis matrix has rank $K = \tilde{K} + 1$ (in our work, $K = 25$).

REFERENCES

[1] Accessed: Aug. 19, 2020. [Online]. Available: http://www.headwallphotonics.com

[2] Accessed: Aug. 19, 2020. [Online]. Available: http://www.hyspex.com

[3] Accessed: Aug. 19, 2020. [Online]. Available: http://www.itres.com

[4] J. R. Schott, *Remote Sensing: The Image Chain Approach*, 2nd ed. New York, NY, USA: Oxford Univ. Press, 2007.

[5] M. K. Griffin and H. K. Burke, "Compensation of hyperspectral data for atmospheric effects," *Lincolm Lab. J.*, vol. 14, no. 1, pp. 29–54, 2003.

[6] B.-C. Gao, K. B. Heidebrecht, and A. F. H. Goetz, "Derivation of scaled surface reflectances from AVIRIS data," *Remote Sens. Environ.*, vol. 44, nos. 2–3, pp. 165–178, May 1993.

[7] F. A. Kruse, "Comparison of ATREM, ACORN, and FLAASH atmospheric corrections using low-altitude AVIRIS data of Boulder, CO," in *Proc. 13th JPL Airborne Geoscience Workshop*, Pasadena, CA, USA, 2004, pp. 1–2.

[8] R. Richter and D. Schläpfer, "Geo-atmospheric processing of airborne imaging spectrometry data. Part 2: Atmospheric/topographic correction," *Int. J. Remote Sens.*, vol. 23, no. 13, pp. 2631–2649, Jan. 2002.

[9] N. Acito and M. Diani, "Unsupervised Atmospheric Compensation of airborne hyperspectral images in the VNIR spectral range," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 7, pp. 3924–3940, 2018.

[10] E. J. Ientilucci and S. Adler-Golden, "Atmospheric compensation of hyperspectral data: An overview and review of in-scene and physics-based approaches," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 31–50, Jun. 2019.

[11] N. Acito and M. Diani, "Atmospheric column water vapor retrieval from hyperspectral VNIR data based on low-rank subspace projection," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 7, pp. 3924–3940, Jul. 2018.

[12] P. S. Chavez, "An improved dark-object subtraction technique for atmospheric scattering correction of multispectral data," *Remote Sens. Environ.*, vol. 24, no. 3, pp. 459–479, Apr. 1988.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:34:35 UTC from IEEE Xplore. Restrictions apply.

[13] F. A. Kruse, "Use of airborne imaging spectrometer data to map minerals associated with hydrothermally altered rocks in the Northern Grapevine mountains, Nevada, and California," *Remote Sens. Environ.*, vol. 24, no. 1, pp. 31–51, Feb. 1988.
[14] L. Bernstein, X. Jin, B. Gregor, and S. Adler-Golden, "The quick atmospheric correction (QUAC)," in *Proc. 4th Workshop Hyperspectral Image Signal Process.*, Shangai, China, 2012, pp. 1–8.
[15] C. J. Miller, "Performance assessment of ACORN atmospheric correction algorithm," in *Proc. Algorithms Technol. Multispectral, Hyperspectral, Ultraspectral Imag.*, Orlando, FL, USA, Aug. 2002, pp. 438–449.
[16] S. M. Adler-Golden *et al.*, "FLAASH, A MODTRAN4 atmospheric correction package for hyperspectral data retrievals and simulations," in *Proc. 7th Ann. JPL Airborne Earth Sci. Workshop*, Pasadena, CA, USA, 1998, pp. 9–14.
[17] R. de los Reyes *et al.*, "PACO: Python-based atmospheric correction," *Sensors*, vol. 20, no. 5, p. 1428, Mar. 2020.
[18] D. R. Thompson *et al.*, "Optimal estimation of spectral surface reflectance in challenging atmospheres," *Remote Sens. Environ.*, vol. 232, pp. 1–15, Oct. 2019.
[19] A. Berk, L. S. Bernstein, and D. C. Robertson, *MODTRAN: A Moderate Resolution Model for LOWTRAN 7*. Burlington, MA, USA: Spectral Sciences, 1989.
[20] R. Richter, D. Schläpfer, and A. Müller, "An automatic atmospheric correction algorithm for visible/NIR imagery," *Int. J. Remote Sens.*, vol. 27, no. 10, pp. 2077–2085, May 2006.
[21] D. Schläpfer, C. C. Borel, J. Keller, and K. I. Itten, "Atmospheric precorrected differential absorption technique to retrieve columnar water vapor," *Remote Sens. Environ.*, vol. 65, no. 3, pp. 353–366, Sep. 1998.
[22] N. Acito, M. Diani, and G. Corsini, "CWV-Net: A deep neural network for atmospheric column water vapor retrieval from hyperspectral VNIR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 1–17, Nov. 2020.
[23] N. Acito, M. Diani, and G. Corsini, "Coupled subspace-based atmospheric compensation of LWIR hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5224–5238, Aug. 2019.
[24] N. Westing, K. C. Gross, B. J. Borghetti, J. Martin, and J. Meola, "Learning set representations for LWIR in-scene atmospheric compensation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 1438–1449, Apr. 2020.
[25] S. M. Adler-Golden, M. W. Matthew, A. Berk, M. J. Fox, J. Lee, and A. J. Ratkowski, "Improvements in aerosol retrieval for atmospheric correction," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Boston, MA, USA, Jul. 2008, pp. 130–133.
[26] A. A. Semenov, A. V. Moshkov, V. N. Pozhidayev, A. Barducci, P. Marcoionni, and I. Pippi, "Estimation of normalized atmospheric point spread function and restoration of remotely sensed images," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 7, pp. 2623–2634, Jul. 2011.
[27] D. Manolakis, R. Lockwood, and T. Codey, *Hyperspectral Imaging Remote Sensing*. Cambridge, U.K.: Cambridge Univ. Press, 2016.
[28] D. R. Thompson *et al.*, "Retrieval of atmospheric parameters and surface reflectance from visible and shortwave infrared imaging spectroscopy data," *Surv. Geophys.*, vol. 40, no. 3, pp. 333–360, May 2019.
[29] D. Schläpfer and R. Richter, "Spectral polishing of high resolution imaging spectroscopy data," *Proc. 7th SIG-IS Workshop Imag. Spectrosc.*, Edinburgh, U.K., 2018, p. 7. [Online]. Available: http://www.daniel-schlaepfer.ch/pdf/Schla epfer IS2011_polis h.pdf
[30] N. Acito, M. Diani, and G. Corsini, "Subspace-based temperature and emissivity separation algorithms in LWIR hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 3, pp. 1523–1537, Mar. 2019.
[31] A. M. Baldridge, S. J. Hook, C. I. Grove, and G. Rivera, "The ASTER spectral library version 2.0," *Remote Sens. Environ.*, vol. 113, no. 4, pp. 711–715, Apr. 2009.
[32] R. N. Clark *et al.*, "USGS digital spectral library splib06a: U.S. Geological Survey Data Series 231," U.S. Geological Survey, Reston, VA, USA, Tech. Rep. 231, 2007, doi: 10.3133/ds231.
[33] Accessed: Aug. 19, 2020. [Online]. Available: http://opticleaf.ipgp.fr/index.php?page
[34] T. Skauli, "Sensor noise informed representation of hyperspectral data, with benefits for image storage and processing," *Opt. Express*, vol. 19, no. 14, pp. 13031–13046, Jul. 2011.
[35] N. Acito, M. Diani, and G. Corsini, "Signal-dependent noise modeling and model parameter estimation in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 8, pp. 2957–2971, Aug. 2011.
[36] N. Acito, S. Matteoli, A. Rossi, M. Diani, and G. Corsini, "Hyperspectral airborne 'Viareggio 2013 Trial' data collection for detection algorithm assessment," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 6, pp. 2365–2376, Oct. 2016.
[37] A. M. Giannandrea *et al.*, "The SHARE 2012 data campaign," *Proc. SPIE*, vol. 8743, May 2013, Art. no. 87430F.
[38] B.-C. Gao, M. J. Montes, C. O. Davis, and A. F. H. Goetz, "Atmospheric correction algorithms for hyperspectral remote sensing data of land and ocean," *Remote Sens. Environ.*, vol. 113, pp. 17–24, Sep. 2009.
[39] W. C. Therrien, *Discrete Random Signals and Statistical Signal Processing*. Upper Saddle River, NJ, USA: Prentice-Hall, 1992.



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